POWER (Pi +5V -> charge-pump -5V -> +/-2.5V DAC reference)

2 x 8 buffers paralleled: half the 16 ohm output resistance of one package, ~0.6 V more rail (10b)

Reservoir sits AT the DAC gate: the DAC's mean supply current is proportional to the signal, so this rail must be LOW IMPEDANCE, not filtered by a series R.

192 kHz pump drive = 4x the 48 kHz output rate, so residual ripple lands on a CIC decimator null.

REFERENCE (+/-2.5V, ratiometric -- tracks +5V so rail noise cancels as gain, not additive noise)

ANALOG ISLAND +5VA (10 uH + 470 u, fc 2.3 kHz)

~35 dB down at the pump's 192 kHz. Feeds the op-amps, the comparators' bias, the DAC gates and the reference, so the reference stays ratiometric to the gates' own rail. Both chokes are the shop's 10u 3A part; L1 is the inlet from the Pi. Neither is a resistor: DCR ~ 0.05 R keeps the DAC reference stiff (design-notes 6).

HAT ID EEPROM (optional; 24LC32 DIP-8, ORDER)

J6 open = write-protected (normal). Short J6 to program the overlay with eepflash.sh; then open it again.

BUTTONS (to the Pi)

SHUTDOWN / WAKE GPIO3

USER GPIO23

CLOCK AND DIGITAL (6.144 MHz -> /2 BCLK 3.072M, /4 MCLK 1.536M, /128 LRCLK 48k; L and R interleaved onto one DIN)

Jumper 1-2 = crystal Pierce (ps jitter -> ~100 dB floor).
Jumper 2-3 = Pi GPCLK0: ~1 ns -> 68 dB floor, the dominant noise source if left there. Bring-up only.

Q0=BCLK 3.072M Q1=MCLK 1.536M
Q4=PUMP 192k Q6=LRCLK 48k

MCLK selects: DIN carries R,L,R,L... at 3.072 Mbps.
64 BCLK per LRCLK frame = 32 L bits + 32 R bits
= exactly one OSR-32 sample per channel.

U3B buffers LRCLK for the clock-alive LED; other spares tied low

LEVEL SHIFT to 3.3 V. 74HC4049 tolerates inputs above its own VCC, which is exactly what makes it legal here. Two inverters per signal so BCLK polarity is preserved -- inverting BCLK would make the Pi sample on the data transition.

MCLK SOURCE DAMPING

Divider Q1 -> 33R2 -> MCLK to both retiming flip-flops and the interleave mux. Series damping at the one source; the edge is the jitter-critical one, so the value stays small.

RASPBERRY PI 4 on a 2x20 socket, mounted UNDER the board.
1/17=3V3 2/4=5V in (through FB1 onto +5V) 7=GPCLK0 GPIO4
12=BCLK GPIO18 35=LRCLK GPIO19 38=DIN GPIO20 even GND
Pi is I2S SLAVE: this board is the clock master.

STATUS LEDS (front edge; CLK and CLIP L/R are with their detectors)

CLOCK ALIVE (diode pump off LRCLK: on only while it toggles)

CHANNEL L – 3rd-order CT delta-sigma, fs 1.536 MHz, OSR 32

INTEGRATOR 1 (a1 = 0.247)

INTEGRATOR 2 (a2 = 0.321, + resonator)

INTEGRATOR 3 (a3 = 0.611)

RESONATOR INVERTER (g = 0.0297)

2u2 + 47k -> 1.5 Hz HP; 1k + 1n5 -> 106 kHz anti-alias.
Trim sets full scale: 3.5 Vpk (2.5 Vrms) at wiper max, and only attenuates from there. Overload is 0.70 FS = 1.7 Vrms in.
For a quieter source drop R21: FS = 2.5 V x R21/R22.

QUANTISER (LM311 + retiming D-FF)
R34 is the excess-loop-delay compensation: without it the LM311's 200 ns costs 8 dB and most of the overload margin.
R36 = R33 is what centres the threshold on VREF_P.

CLIP THRESHOLDS (+/-2.0 V from the references) and the LM339's supply

TH_P = +2.47 x 100/122.1 = +2.02 V
TH_N = -2.43 x 100/122.1 = -1.99 V

CLIP L (window on INT1_L against TH_P / TH_N)

1-BIT DAC DRIVE
DACN = /Q, DACP = Q (taken from Q and /Q so the two edges are as close to simultaneous as the parts allow)

TEST POINTS L (odd pins GND)

TONEARM / CHASSIS GROUND (J4 shorted = hard bond; open = lift network)

D15/D16 conduct CHASSIS <-> GND above 0.7 V either way; 10R0 carries hum current, 100n carries RF. Enclosure post and the turntable's spade both land on J3.

TEST POINTS, DIGITAL AND REFERENCE (odd pins GND)

CHANNEL R – 3rd-order CT delta-sigma, fs 1.536 MHz, OSR 32

INTEGRATOR 1 (a1 = 0.247)

INTEGRATOR 2 (a2 = 0.321, + resonator)

INTEGRATOR 3 (a3 = 0.611)

RESONATOR INVERTER (g = 0.0297)

2u2 + 47k -> 1.5 Hz HP; 1k + 1n5 -> 106 kHz anti-alias.
Trim sets full scale: 3.5 Vpk (2.5 Vrms) at wiper max, and only attenuates from there. Overload is 0.70 FS = 1.7 Vrms in.
For a quieter source drop R61: FS = 2.5 V x R61/R62.

QUANTISER (LM311 + retiming D-FF)
R74 is the excess-loop-delay compensation: without it the LM311's 200 ns costs 8 dB and most of the overload margin.
R76 = R73 is what centres the threshold on VREF_P.

SPICE TESTBENCHES (sim/, one project each -- click to open)

A rails and the -5 V charge pump
B the +/-2.5 V ratiometric reference
C clock buffer and 74HC4040 divider
D integrator 1 with a real TL074
E LM311 quantiser on a single supply
F the 1-bit DAC and its offset leg
G retiming, interleave and the 3.3 V level shift
H one complete channel, with the click test

CLIP R (window on INT1_R against TH_P / TH_N)

1-BIT DAC DRIVE
DACN = /Q, DACP = Q (taken from Q and /Q so the two edges are as close to simultaneous as the parts allow)

TEST POINTS R (odd pins GND)

Sheet: /
File: vinyl_adc_rev_d.kicad_sch
Title: Discrete 3rd-order delta-sigma vinyl ADC – stereo, 48 kHz – REV
Size: A0
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 1/1